

**THIS PRESENTATION MUST BE UPLOADED
BEFORE THE PRESENTATION TIME (14:15)**

Galactic Speed Project

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The Bailey algorithm

- Variation of the Cooley–Tukey FFT algorithm originally designed for systems with hierarchical memory
- Could provide a good starting point for optimizations
 - Lots of parallelization
 - Cache efficiency
 - Row/Col major alternance very efficient in hardware (BRAM)
- Off-core computation, multiple small FFT that can be loaded in memory

1

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Arrange input data sequence

2

F	F	F	F
F	F	F	F
T	T	T	T
1	2	3	4

First set of FFTs

3 $\times e^{\pm 2\pi ijk/16}$ Correction multiplication

4

F	F	T	1
F	F	T	2
F	F	T	3
F	F	T	4

Second set of FFTs

1	5	9	13
2	6	10	14
3	7	11	15
4	8	12	16

Result sequence

Fixed Point

Goal:

- Reduce resource usage
- Improve SNR since FXP is computed the same way on the pynq and on the core i9

How:

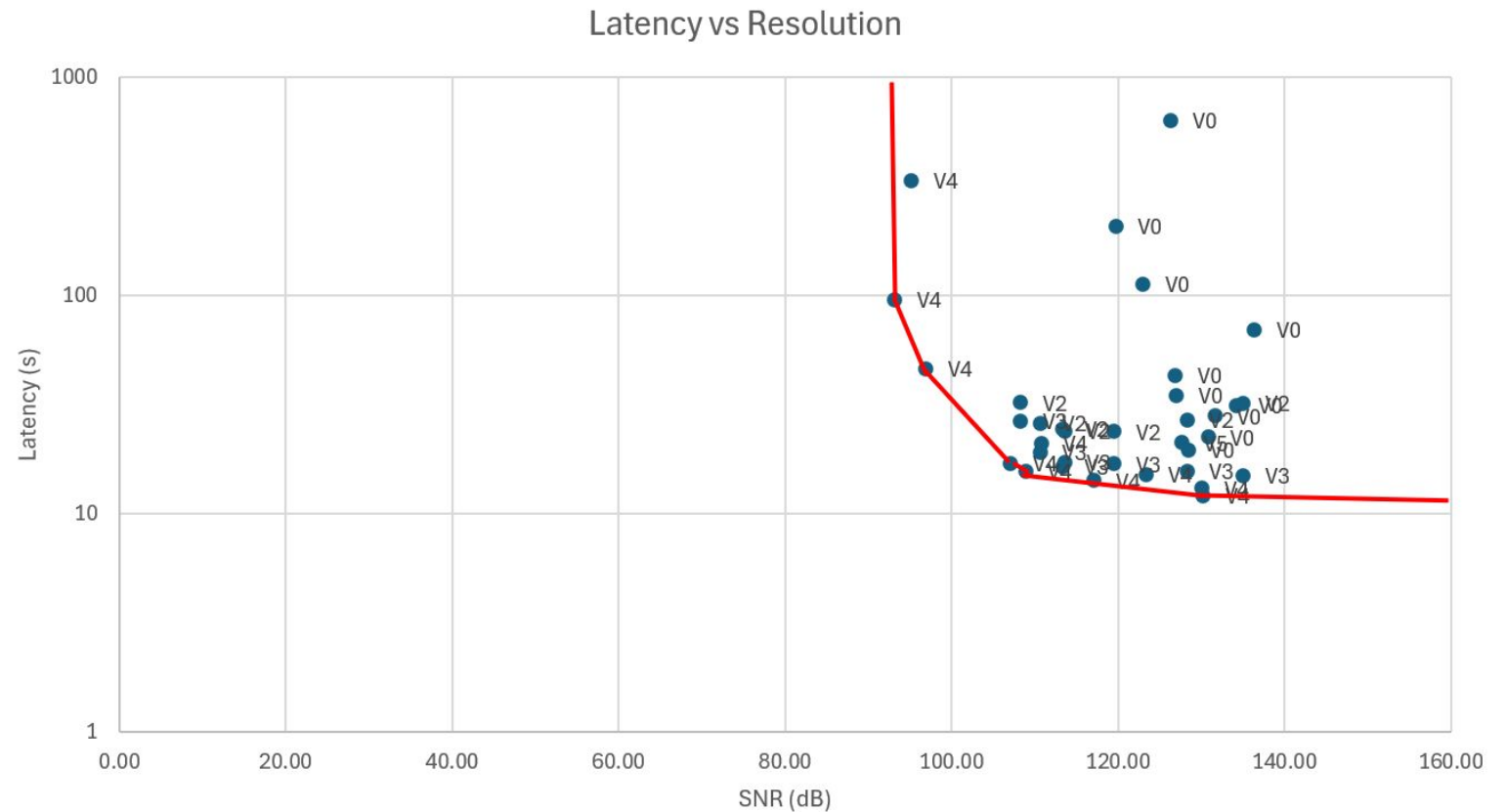
- Normalized the FFT by nfft to stay in the same range for all configurations and stages

Issue:

- We implemented the FXP only in our FFT so we do lots of FXP/FP conversion

Summary of the project

- Number of solutions generated: 4
 - V2: Bailey software-hardware
 - V3: Bailey software-hardware opti
 - V4: Bailey full software
 - V5: Bailey full hardware

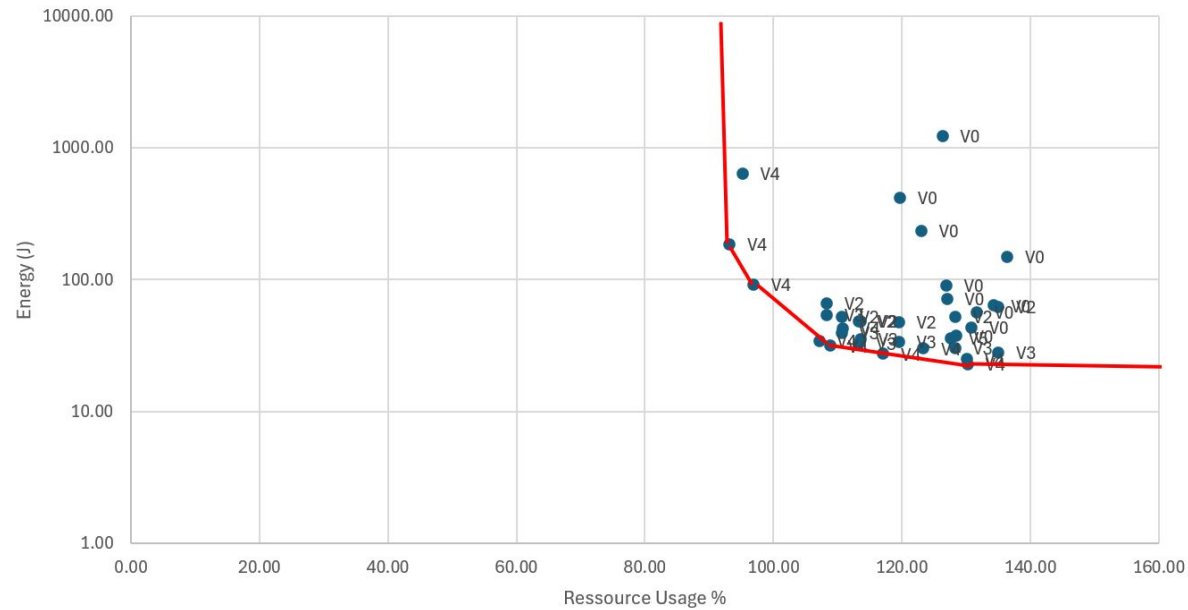




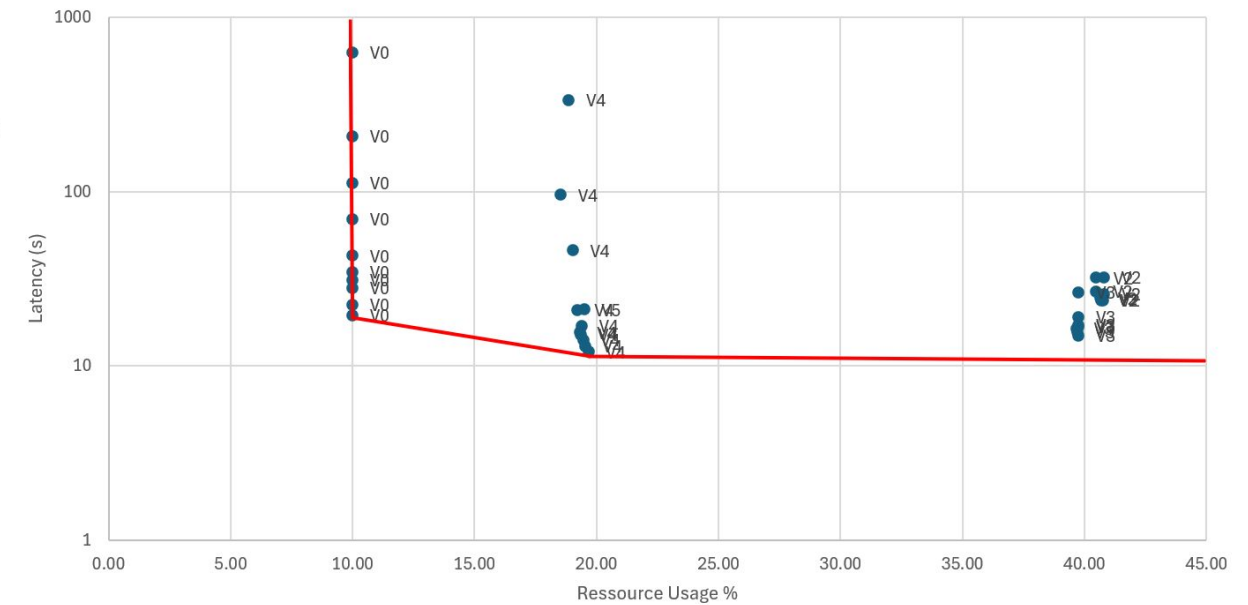
Name	Config	Description	Time FFT (s)	Time tot (s)	Frequency (MHz)	%CoreUtilization	LUTs	FFs	BRAMs	DSPs	Cost	Energy FFT	Energy (J)	SNR (dB)	Pareto
V0	0	Base-software	14.414	19.425	-	50.00	0	0	0	0	10.00	27.70	37.39	128.40	Yes
V0	1	Base-software	17.061	22.336	-	50.00	0	0	0	0	10.00	33.41	43.30	130.80	Yes
V0	2	Base-software	22.445	27.883	-	50.00	0	0	0	0	10.00	45.72	56.67	131.60	Yes
V0	3	Base-software	25.117	31.175	-	50.00	0	0	0	0	10.00	51.64	63.68	134.20	Yes
V0	4	Base-software	28.215	34.552	-	50.00	0	0	0	0	10.00	58.53	71.21	127.00	Yes
V0	5	Base-software	35.088	42.815	-	50.00	0	0	0	0	10.00	74.90	90.32	126.90	Yes
V0	6	Base-software	55.651	69.326	-	50.00	0	0	0	0	10.00	122.08	149.00	136.40	Yes
V0	7	Base-software	62.56	112.198	-	50.00	0	0	0	0	10.00	138.72	235.40	123.00	Yes
V0	8	Base-software	69.659	206.951	-	50.00	0	0	0	0	10.00	151.81	416.37	119.70	Yes
V0	9	Base-software	72.865	630.648	-	50.00	0	0	0	0	10.00	158.82	1231.27	126.30	Yes
V2	0	Bailey software-hardware	27.281	32.025	100	48.33	17766	15936	66.5	128	40.48	52.80	61.93	135.02	No
V2	1	Bailey software-hardware	21.710	26.818	100	48.23	17766	15936	66.5	128	40.46	42.98	52.11	128.27	No
V2	2	Bailey software-hardware	18.921	24.489	100	49.20	17766	15936	66.5	128	40.65	37.50	48.56	113.27	No
V2	3	Bailey software-hardware	17.624	23.683	100	49.73	17766	15936	66.5	128	40.76	35.27	47.53	119.51	No
V2	4	Bailey software-hardware	17.424	23.724	100	49.36	17766	15936	66.5	128	40.68	36.01	48.45	113.54	No
V2	5	Bailey software-hardware	17.845	25.768	100	49.92	17766	15936	66.5	128	40.79	36.78	52.57	110.61	No
V2	6	Bailey software-hardware	19.469	32.166	100	49.96	17766	15936	66.5	128	40.80	40.34	66.21	108.29	No
V3	0	Bailey software-hardware with opti	9.876	14.841	100	50	15397	14691	67	128	39.76	18.52	28.16	135.02	No
V3	1	Bailey software-hardware with opti	10.359	15.569	100	49.79	15397	14691	67	128	39.72	20.10	30.10	128.27	No
V3	2	Bailey software-hardware with opti	10.951	16.406	100	49.565	15397	14691	67	128	39.67	22.03	32.62	113.27	No
V3	3	Bailey software-hardware with opti	11.159	16.825	100	50	15397	14691	67	128	39.76	22.22	33.84	119.51	No
V3	4	Bailey software-hardware with opti	11.167	17.079	100	50	15397	14691	67	128	39.76	23.23	35.26	113.54	No
V3	5	Bailey software-hardware with opti	11.566	19.06	100	50	15397	14691	67	128	39.76	24.06	39.37	110.61	No
V3	6	Bailey software-hardware with opti	12.874	26.38	100	50	15397	14691	67	128	39.76	27.04	54.06	108.29	No
V4	0	Bailey full software	8.764	12.001	-	98.35	0	0	0	0	19.67	16.27	22.77	130.23	Yes
V4	1	Bailey full software	9.643	12.998	-	97.68	0	0	0	0	19.54	18.23	24.90	130.11	Yes
V4	2	Bailey full software	10.685	14.117	-	97.25	0	0	0	0	19.45	20.72	27.37	117.09	Yes
V4	3	Bailey full software	11.094	15.038	-	96.85	0	0	0	0	19.37	21.93	30.08	123.34	Yes
V4	4	Bailey full software	11.249	15.535	-	96.54	0	0	0	0	19.31	23.11	31.53	108.86	Yes
V4	5	Bailey full software	11.906	16.923	-	96.94	0	0	0	0	19.39	24.47	34.54	107.10	Yes
V4	6	Bailey full software	13.001	20.968	-	96.02	0	0	0	0	19.20	26.80	42.64	110.73	Yes
V4	7	Bailey full software	16.821	45.961	-	95.13	0	0	0	0	19.03	34.84	91.41	96.89	Yes
V4	8	Bailey full software	18.421	95.792	-	92.67	0	0	0	0	18.53	38.10	185.77	93.17	Yes
V4	9	Bailey full software	18.068	334.26	-	94.33	0	0	0	0	18.87	37.31	638.70	95.18	Yes
V5	0	Bailey full hardware	16.084	21.04	100	49.945	5797	7348	11	48	19.48	27.68	35.82	127.60	No

Summary of the project

Energy vs Resolution



Latency vs Resources



Solution V2: First Bailey software-hardware

Name	Config	Description	Time FFT (s)	Time tot (s)	Frequency (MHz)	%CoreUtilization	LUTs	FFs	BRAMs	DSPs	Cost	Energy FFT	Energy (J)	SNR (dB)
V2	0	Bailey software-hardware	27.281	32.025	100	48.33	17766	15936	66.5	128	40.48	52.80	61.93	135.02
V2	1	Bailey software-hardware	21.710	26.818	100	48.23	17766	15936	66.5	128	40.46	42.98	52.11	128.27
V2	2	Bailey software-hardware	18.921	24.489	100	49.20	17766	15936	66.5	128	40.65	37.50	48.56	113.27
V2	3	Bailey software-hardware	17.624	23.683	100	49.73	17766	15936	66.5	128	40.76	35.27	47.53	119.51
V2	4	Bailey software-hardware	17.424	23.724	100	49.36	17766	15936	66.5	128	40.68	36.01	48.45	113.54
V2	5	Bailey software-hardware	17.845	25.768	100	49.92	17766	15936	66.5	128	40.79	36.78	52.57	110.61
V2	6	Bailey software-hardware	19.469	32.166	100	49.96	17766	15936	66.5	128	40.80	40.34	66.21	108.29

- Only the FFT inside hardware -> 2^{10} NFFT maximum
- Use of caching with ACP port
- Twiddles in LUT
- Ping pong buffer
- Dataflow with 8 workers
- Test with kernel -> slower because the switch between user-space and kernel-space is costly

Solution V3: Bailey software-hardware optimized

Name	Config	Description	Time FFT (s)	Time tot (s)	Frequency (MHz)	%CoreUtilization	LUTs	FFs	BRAMs	DSPs	Cost	Energy FFT	Energy (J)	SNR (dB)
V3	0	Bailey software-hardware with opti	9.876	14.841	100	50	15397	14691	67	128	39.76	18.52	28.16	135.02
V3	1	Bailey software-hardware with opti	10.359	15.569	100	49.79	15397	14691	67	128	39.72	20.10	30.10	128.27
V3	2	Bailey software-hardware with opti	10.951	16.406	100	49.565	15397	14691	67	128	39.67	22.03	32.62	113.27
V3	3	Bailey software-hardware with opti	11.159	16.825	100	50	15397	14691	67	128	39.76	22.22	33.84	119.51
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V3	5	Bailey software-hardware with opti	11.566	19.06	100	50	15397	14691	67	128	39.76	24.06	39.37	110.61
V3	6	Bailey software-hardware with opti	12.874	26.38	100	50	15397	14691	67	128	39.76	27.04	54.06	108.29

- 2 AXI bus to burst real + imag
- Software optimizations

Solution V3: Bailey software-hardware optimized

```

hls_thread_local hls::stream<TDataIn> input_streams[NUM_WORKERS];
#pragma HLS STREAM variable=input_streams depth=MAX_NFFT
hls_thread_local hls::stream<TDataOut> output_streams[NUM_WORKERS];
#pragma HLS STREAM variable=output_streams depth=MAX_NFFT

hls_thread_local hls::stream<TDataConfig> config_streams[NUM_WORKERS];
#pragma HLS STREAM variable=output_streams depth=1

#pragma HLS dataflow

ReadInputs(log2_nfft, numHW, In_real, In_imag, input_streams, config_streams);

hls_thread_local hls::task worker0(ComputeFFT, config_streams[0], input_streams[0], output_streams[0]);
#ifdef NUM_WORKERS_2
hls_thread_local hls::task worker1(ComputeFFT, config_streams[1], input_streams[1], output_streams[1]);
#endif
#ifdef NUM_WORKERS_4
hls_thread_local hls::task worker2(ComputeFFT, config_streams[2], input_streams[2], output_streams[2]);
hls_thread_local hls::task worker3(ComputeFFT, config_streams[3], input_streams[3], output_streams[3]);
#endif
#ifdef NUM_WORKERS_8
hls_thread_local hls::task worker4(ComputeFFT, config_streams[4], input_streams[4], output_streams[4]);
hls_thread_local hls::task worker5(ComputeFFT, config_streams[5], input_streams[5], output_streams[5]);
hls_thread_local hls::task worker6(ComputeFFT, config_streams[6], input_streams[6], output_streams[6]);
hls_thread_local hls::task worker7(ComputeFFT, config_streams[7], input_streams[7], output_streams[7]);
#endif

WriteOutputs(numHW, log2_nfft, Out_real, Out_imag, output_streams);

```

Solution V4: Bailey full software

Name	Config	Description	Time FFT (s)	Time tot (s)	Frequency (MHz)	%CoreUtilization	LUTs	FFs	BRAMs	DSPs	Cost	Energy FFT	Energy (J)	SNR (dB)
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V4	6	Bailey full software	13.001	20.968	-	96.02	0	0	0	0	19.20	26.80	42.64	110.73
V4	7	Bailey full software	16.821	45.961	-	95.13	0	0	0	0	19.03	34.84	91.41	96.89
V4	8	Bailey full software	18.421	95.792	-	92.67	0	0	0	0	18.53	38.10	185.77	93.17
V4	9	Bailey full software	18.068	334.26	-	94.33	0	0	0	0	18.87	37.31	638.70	95.18

- Usage of OpenMP to parallelize with 2 threads

Solution V5: Bailey full hardware

Name	Config	Description	Time FFT (s)	Time tot (s)	Frequency (MHz)	%CoreUtilization	LUTs	FFs	BRAMs	DSPs	Cost	Energy FFT	Energy (J)	SNR (dB)
V5	0	Bailey full hardware	16.084	21.04	100	49.945	5797	7348	11	48	19.48	27.68	35.82	127.60

- Use ap_uint to reduce resource usage
- Highly specific to config 0
- Did not had time to optimize it (no unrolling, poor burst...)

- Major achievements of your project
 - Dataflow
 - Bailey FFT implementation
 - Cache optimization
 - Good scaling
 - “High flexibility”
- Hurdles encountered during the project
 - Cache decoherence
 - Kernel crash
 - Strategic bottlenecks
- Final words about the project and the course
 - Exploring many technical options and being highly creative is exciting, but it also makes it easy to head in the wrong direction.
 - Incredibly time-consuming but super interesting.